Ananya Panayappan

IBDP Student · 2027
ananya.life.22@gmail.com | +91 701 951 0673 | ananyapanayappan.com | www.linkedin.com/in/ananya-panayappan/

EDUCATION

IB Diploma Programme

Candor International School, Bengaluru

Expected June 2027

- IBDP1 43/45, with 7s in Math AA HL, Economics HL, and Computer Science HL.

Mathematics Extended Essay

- Analysed the mathematical underpinnings, assumptions and limitations of Geometric Brownian Motion (GBM) and its validity with regards to S&P 500 stock price data.

Economics Internal Assessment

- Applied economic theory to evaluate real-world issues and policies including Pigouvian taxation, imported inflation, and trade protection.

IGCSE Distinction

Candor International School · June 2025

- School topper: 1st of 60, with eight A* grades.
- 98 PUM in Additional Mathematics; 97 PUM in Economics and Computer Science.
- Earned ICE Distinction

STANDARDIZED TESTING

SAT

1570 · May 2026

- 800 in Math
- 770 in Reading and Writing

AP

Microeconomics · May 2026

- Grade of 5

Calculus BC · May 2026

- Grade of 5
- Calculus AB Subscore: 5

RESEARCH & INTERNSHIPS

Pioneer Academics — Student Researcher — Grades 10–11

- Developed a quantitative portfolio-resilience framework examining Indian markets during COVID-19 and demonetisation, focusing on stochastic processes, probability, and quantitative finance.
- Earned an A grade, ranked in the top 5% of undergraduate scholars evaluated by NYU Professor Hesam Oveys.
- Research paper accepted for publication in the International Journal of High School Research, due for publication in November 2026.

Independent Research — Financial Econometrics — Grade 11-Ongoing

- Investigating the relationship between India VIX and macroeconomic drivers through a regression model of global volatility transmission to Indian markets, mentored by an IIMA Professor Vineet Virmani.

Dyota Solutions — Quantitative Finance Intern — Grade 11-12

- Analysed 12 equity fund categories against their benchmarks, simulating lump-sum and SIP returns using CAGR, XIRR, and win rates under the mentorship of CEO Kaushik Ramachandran.

Math Extended Essay — Geometric Brownian Motion — Grade 11-12

- Derived GBM using Itô's Lemma and evaluated its assumptions against five years of S&P 500 data using statistical analysis of log returns, volatility, and overnight price movements.
- Identified heavy-tailed returns and volatility clustering, assessing implications for financial modelling and alternative models such as GARCH and jump-diffusion.

PROJECTS

PocketSmart — Founder & Lead Developer — Grade 11

- Conceptualised, developed, and launched a personal-finance app for teenagers to track allowances, reflect on spending, set savings goals, and evaluate opportunity costs.
- Designed allowance-tracking and automated savings features to encourage more deliberate financial decisions and greater investment
- Recently launched on Google Play Store:

Mitra Jyothi (NGO for Blind) — Volunteer Project Lead — Grades 9-12

- Built Braille validation tool for Braille press, achieving reported accuracy above 70% and easing bottlenecks in accessible textbook production. Saved 2000+ annual proofreading hours.
- Contributed to initiative serving a beneficiary population of 4,000+ students across 77 schools in India.

OptimalHire — AI Recruitment Application — Grades 11-12

- Built recruitment web app with React.js, Python Flask, MySQL, BERT, and Siamese network to rank job/candidate match by semantic similarity.

SKILLS

Mathematics & Statistics: Mathematical modelling, stochastic processes, game theory, graph theory, statistical analysis, regression analysis, calculus, probability theory, combinatorial modelling.

Economics & Econometrics: Data-driven economic research, quantitative economic analysis, econometrics, statistical modelling, microeconomics, macroeconomics.

Computational Methods & Technology: Statistical machine learning, natural language processing, BERT, Siamese networks, React.js, Python, Flask, MySQL.

INTERESTS & INITIATIVES

Pioneer Alumni Network: Engaged with Pioneer Academics students, sharing insights and guidance on navigating the research process through the Pioneer Academics programme.

Meraki Editorial Board: Acted as Chief Editor for school magazine, writing creative, formal, and informative pieces

Other Interests: vocal music, keyboard, original music composition, baking, and pastry techniques.

TECHNICAL QUALIFICATIONS

WhiteHat Jr · 2020
- Certified Game Developer with Deep UI/UX Interface

GiveMeFive · 2022
- Certified Web Developer Level 2

LANGUAGES

English · Tamil · Hindi

LEADERSHIP & ENTREPRENEURSHIP

Mysteraskin — Co-Founder, Finance & Creative Lead Grades 10-12
- Lead financial planning, budgeting, cash-flow management, and analysis while developing product concepts for a handmade skincare venture.
- Website

Project Flow — Founder & Digital Media Coordinator Grades 10-12
- Spearheaded menstrual health education across 22 schools, organised sustainable menstrual product drives, and built the project website.
- Website

Math Club, Candor International School — Founder Grades 11-12
- Founded a math club for Grades 6-9 with applied game theory and graph theory activities; organised schoolwide MegaMatics competition.

CanMUN & Aletheia — Executive Organising Team Grades 10-11
- Helped organise CanMUN (Candor MUN) and Aletheia (interschool sports event); handled registrations and security for 1,000+ attendees.

ADDITIONAL COURSEWORK

Johns Hopkins University Grade 10
- Financial Mathematics, Grade A (GPA 4.0/4.0), concurrent enrollment.

Columbia University Grade 10
- Thinking and Problem Solving: Math in the Real World (graph theory, combinatorial modelling, game theory).

Wharton Global Youth Program Grade 11
- Essentials of Finance (Pitched an investment proposal; worked in teams to perform fundamental valuation of the Costco stock).

Arizona State University (ULC) Grades 9-11
- Grade A in College Algebra, Precalculus, Calculus for Engineers I and II, Statistics, Microeconomics.

Coursera - Specialized Online Coursework Grades 9-12
- Financial Markets (Yale University, Robert Shiller)
- Programming for Everybody Specialisation (University of Michigan)
- The Nuts and Bolts of Machine Learning (Google)
- Enjoyable Econometrics (Erasmus University)
- Singing Popular Music (Berklee College of Music)

HONORS & AWARDS

- SOF International Mathematics Olympiad: International Rank 4 (2021) & Consecutive School Gold Medals (2019-24).
- 1st Place, Epsilon Nationwide Mathematics Competition (2024, 2025).
- Best Rationale Award, Wharton Global Youth Program (2026).
- 1st Place, Eton College: Startup idea elevator pitch presentation (2023).
- 1st Place in Hackathons; Binary Brawl (2023), SNIS TechFest (2023), and Candor TechFest (2024).
- 1st Place, Candor Entrepreneurship Day: Harry Potter soap stall (2023).
- 1st Place, Candor Verbattle: intraschool debate competition (2023).